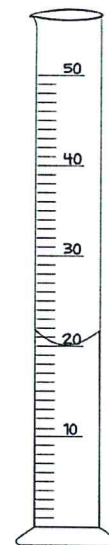


Name: Key

Scientific Method Test Honors

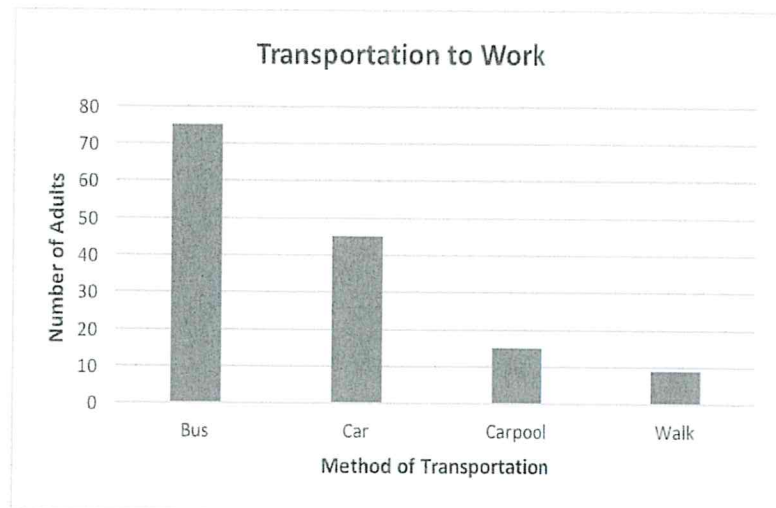
Part 1 Multiple Choice: Write the answer that best answers the question in the blank. (1.5 pts each)

- A 1. What is the part of an experiment that serves as a standard for comparison?
A. Control Group B. Constant
C. Hypothesis D. Variable
- B 2. A testable prediction that predicts a cause and effect relationship between variables is a(n) _____.
A. analysis B. hypothesis
C. observation D. inference
- B 3. Where does the dependent variable go on a graph?
A. X-axis B. Y-axis
C. In the Title D. In the Key
- D 4. Why is it advisable to use tongs when taking a beaker off a laboratory burner?
A. It prevents spilling. B. It prevents the beaker from breaking.
C. It looks cool. D. It allows you to move it if it is too hot.
- D 5. A factor in an experiment that changes in response to the independent variable is the _____.
A. constant B. control group
C. independent variable D. dependent variable
- B 6. Using the graduated cylinder to the right, pick the accurate AND precise measurement below.
A. 22.0 mL B. 20.0 mL
C. 22 mL D. 20 mL
- D 7. If your lab partner's clothes catch on fire, you should first _____.
A. use a fire extinguisher B. panic
C. do nothing. It will burn out eventually. D. tell the teacher
- D 8. All of the following are steps in the scientific method EXCEPT
A. observing and recording data. B. forming a hypothesis.
C. developing a model. D. discarding data that doesn't match the hypothesis.
- B 9. Which of these would be the BEST way to illustrate the percentage of students receiving A's, B's, C's, D's and F's on a recent test?
A. Bar graph B. Circle Graph
C. Line Graph D. Scatter Graph
- A 10. A factor that does NOT change in an experiment is the _____.
A. constant B. control group
C. dependent variable D. independent variable
- D 11. The "cause" of the experiment, what we are purposefully manipulating or changing is the _____.
A. constant B. control group
C. dependent variable D. independent variable
- B 12. What is 78,900,000,000 expressed in scientific notation?
A) 789×10^9 B) 7.89×10^{10}
C) 7.89×10^9 D) 7.89×10^{11}



C 13. Using the graph to the right, how many adults do **NOT** carpool to work?

- A. 15
- B. 50
- C. 130
- D. 160



D 14. The speed of light is approximately 3.0×10^8 m/s. How would this be written in standard notation?

- A) 300,000 m/s
- B) 30,000,000 m/s
- C) 3,000,000 m/s
- D) 300,000,000 m/s

C 15. Mrs. James wants to measure the volume of a liquid very precisely. What is the BEST lab equipment she could use to do this?

- A. beaker
- B. test tube
- C. graduated cylinder
- D. Erlenmeyer flask

A 16. Which of the following is a quantitative observation?

- A. The dog has two legs.
- B. The dog is small.
- C. The dog is cute.
- D. The dog is fuzzy.

C 17. Which type of graph would be MOST appropriate for a track runner to use in order to look at how his speed varied during a 500-meter race?

- A. bar graph
- B. pie graph
- C. line graph
- D. narrative paragraph

A 18. Jenny and Angie are studying how listening to music affects grades. Jenny listened to music when she studied while Angie studied in silence. After two weeks, they took the same math test and Jenny scored higher than Angie. They came to the conclusion that listening to music will improve grades. What comment would you make to Jenny and Angie about their experiment?

- A. They should have tested more people
- B. They should have tested a boy and a girl
- C. This topic has already been studied before
- D. They should have chosen a different subject besides math

D 19. In 1859, Charles Darwin explained a model of why living things change over time. He described this model and the evidence that supported it in a book called *On The Origin of Species*. Which scientific term is used to *describe a testable model that seeks to explain natural phenomena*?

- A. data
- B. hypothesis
- C. law
- D. theory

C 20. A description of how things happen in nature that seems to be true all the time is a _____.

- A. data
- B. hypothesis
- C. law
- D. theory

Part 2 Safety and Equipment: Write the answer for the following questions below. You do not have to use complete sentences. Point values are indicated next to each question.

21. What is the name of the curve a liquid gets when in a graduated cylinder? (1 pt)

meniscus

22. What is the first thing you should always do before starting a lab? (1 pt)

read the instructions

23. What does P.A.S.S. stand for? (2 pts)

Pull Aim Squeeze Sweep

24. Shelley accidentally drops a test tube and it shatters. What two things should she make sure to do? (2 pts)

① tell the teacher ② use broken glass disposal instead of trash can.

25. What should you always do at the end of a lab? (1 pt)

clean up

Part 3 Measurement: Write the answer for the following questions. You do not have to use complete sentences. (1 pt each)

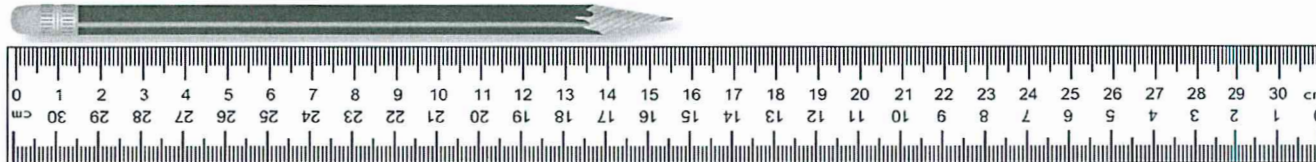
26. What is the SI unit for mass? kg

27. What is the SI unit for length? m

28. What is the SI unit for temperature? K

29. Measure the length of the pencil using the ruler below. Make sure to include units in your answer.

15.50 cm



Part 4 Data Analysis: Use the following data to answer questions #30-33. Point values are indicated next to each question.

Two lab groups measure the length of their notebook four different times. Here is their data:

Group A:

Trial 1	Trial 2	Trial 3	Trial 4
26.5 cm	27.5 cm	27.0 cm	27.7 cm

Group B:

Trial 1	Trial 2	Trial 3	Trial 4
25.5 cm	25.0 cm	23.0 cm	29.5 cm

30. What is the average length of Group A's data? (2 pts)

27.175 cm

31. What is the average length of Group B's data? (2 pts)

25.75 cm

32. Which group's data is more precise? Why? (3 pts)

Group A b/c their data is consistent.

33. If the actual length of the notebook is 26.1 cm, which set of data is more accurate? Why? (3 pts)

Group B b/c the average length they got was 25.75 cm which is closest to the correct value of 26.1 cm.

Part 5 Scientific Method: Read the description of a laboratory investigation below. Use this information to answer questions #34-39. Point values indicated next to each question.

An experiment was designed to investigate the effect of caffeine on the heartbeat of water fleas. Two populations of water fleas were cultured. Both populations had water with the same mineral content, were supplied with identical amounts of bacteria as food, received the same amount of light, and had their temperature maintained at 20°C. Every two hours, water fleas from both populations were selected and their heartbeats were monitored. The fleas of population A had caffeine administered five minutes before their heartbeat was checked. The fleas of population B were given nothing.

34. What is the independent variable in this experiment? (1 pt)

caffeine or not (amt. of caffeine)

35. What is the dependent variable in this experiment? (1 pt)

heartbeat

36. Write a hypothesis, in the correct format, for this experiment. (2 pts)

If the amount of caffeine ↑, then the heartbeat will ↑.

37. What is the control group in this experiment? (1 pt)

Population B

38. What is the experimental group in this experiment? (1 pt)

Population A

39. List two constants in this experiment. (2 pts)

= same mineral content in water, same amt. of food, same amt. of light, same temp., heartbeat measured at same time.

Part 6 Conversions: Convert the following measurements. Make sure to show all work and circle/box your answer so that it stands out. Point values indicated next to each question.

40. 45 kg to g (2 pts)

$$\frac{45 \text{ kg} \times 1000 \text{ g}}{1 \text{ kg}} = \boxed{45,000 \text{ g}}$$

41. 2,570 cm to hm (3 pts)

$$\frac{2570 \text{ cm}}{100 \text{ cm}} \times \frac{1 \text{ m}}{100 \text{ m}} \times \frac{1 \text{ hm}}{10,000 \text{ m}} = \frac{2570}{10,000} = \boxed{0.257 \text{ hm}}$$

42. 58°C to K (3 pts)

$$\begin{array}{l|l} K=? & K = 58 + 273 \\ C = 58^\circ & K = \boxed{331 \text{ K}} \\ K = C + 273 & \end{array}$$

43. 29°C to F (3 pts)

$$\begin{array}{l|l} F=? & F = \left(\frac{9}{5} \times 29\right) + 32 \\ C = 29^\circ & F = 52.2 + 32 \\ F = \left(\frac{9}{5} \times C\right) + 32 & F = \boxed{84.2^\circ \text{F}} \end{array}$$

44. 251 K to °F (4 pts)

$$\begin{array}{l|l|l|l} F=? & K = C + 273 & C = 251 - 273 & F = \left(\frac{9}{5} \times -22\right) + 32 \\ C=? & -273 & -273 & F = -39.6 + 32 \\ K = 251 & C = K - 273 & C = -22^\circ \text{C} & F = \boxed{-7.6^\circ \text{F}} \end{array}$$

44. 3 miles to feet (4 pts)

$$\frac{3 \text{ mi}}{1 \text{ mi}} \times \frac{5,280 \text{ ft.}}{1 \text{ mi}} = \boxed{15,840 \text{ ft.}}$$

45. 0.61 tons to grams (4 pts)

$$\frac{0.61 \text{ tons}}{1 \text{ ton}} \times \frac{2,000 \text{ lbs}}{2.2 \text{ lbs}} \times \frac{1 \text{ kg}}{1 \text{ kg}} \times \frac{1,000 \text{ g}}{1 \text{ kg}} = \frac{1,220,000}{2.2} = \boxed{554,545 \text{ g}}$$

Part 7 Scientific Notation: Convert the following numbers into scientific notation. Make sure to show all work and circle/box your answer so that it stands out. (2 pts each)

46. 2,075,000 2.075×10^6

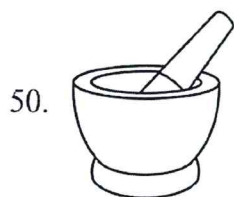
47. 0.0041 4.1×10^{-3}

Part 8 Standard Notation: Convert the following numbers into standard notation. Make sure to show all work and circle/box your answer so that it stands out. (2 pts each)

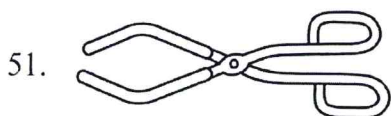
48. 1.17×10^{-4} 0.000117

49. 9.081×10^7 $90,081,000$

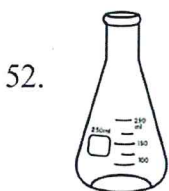
Part 9 Lab Equipment: Name the following lab equipment below. Then write what it is used for. (2 pts each)



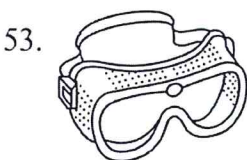
Mortar + pestle = grinding solids into powders



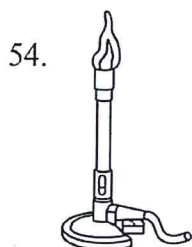
Beaker tongs = carrying hot beakers



Erlenmeyer flask = heating, mixing + measuring chemicals



Safety goggles = protect eyes from chemicals + fire.



Bunsen burner = heating; powered by gas

Bonus: Use the line graph to the right to answer the questions below.

a. Find the number of grams of sugar that will dissolve in 100 grams of water at 80°C.

400g

b. What is the independent variable in the graph?

temperature

c. Find the temperature, in Kelvin, that will dissolve 400 g of sugar in 100 g of water.

$K = ?$

$C = 80$

$K = C + 273$

$K = 80 + 273$

$K = \boxed{353 \text{ K}}$

